

```
$ fswebcam example.jpg
```

An image by the name example.jpg will get saved in the home directory. If the resolution of this image is not up to the mark, change it by using -r option in fswebcam. One example of 1280x720 resolution capturing is shown below. Set this according to your webcam.

```
$ fswebcam -r 1280x720 example.jpg
```

To install sound on Raspbian, install also sound utilities using the command below:

```
$ sudo apt-get install alsa-utils
```

Edit the modules file at /etc/modules using nano editor.

```
$ sudo nano /etc/modules
```

Add the line snd_bcm2835. If snd_bcm2835 is already present, leave the file as it is.

Then, save the file by clicking ctrl+o and exit with ctrl+x.

Now, install mplayer audio movie player using the command:

```
$ sudo apt-get install mplayer
```

Once you have completed all the steps mentioned above, install Festival text-to-speech software using the command:

```
$ sudo apt-get install festival
```

You may try Festival installation using the command below in the terminal and you will hear Hello EFY in the earphones.

```
$ echo "Hello EFY" | festival -tts
```

Once all the above software are installed, copy see.py Python code, which is included in this month's DVD available with Electronics For You Plus version of the magazine in Home folder.

Run see.py by issuing the following command:

```
$ sudo python see.py
```

see.py runs indefinitely to get input from the user.

Note. If the resolution of your camera is not good, OCR performance will be poor and the speech output will also degrade.

We have used Logitech C270 camera for testing this project. The camera resolution by default is 720x340, which is

the maximum resolution supported by this webcam. If the camera is unable to capture the text properly, you will either hear distorted sounds from the speaker or no sound at all.

The text image (example.jpg) being captured by this camera during testing is shown in Fig. 3.

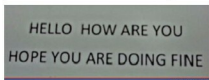


Fig. 3: Text image (example.jpg) captured by the camera during testing

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at source.efymag.com

You can find example.jpg and speech.txt files under Home directory. ●

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